When Treasuries Crowd the Debt Market: Treasury Market Inelasticity and the Basis Trade

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Motivation

Since the Global Financial Crisis,

- the Fed purchased $\approx 40\%$ of long-term Treasuries through quantitative easing (QE).
- more recently, it has offloaded these holdings via quantitative tightening (QT).

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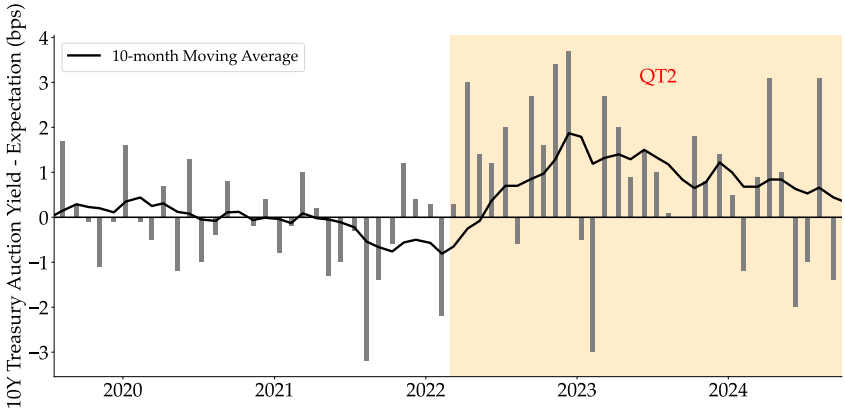
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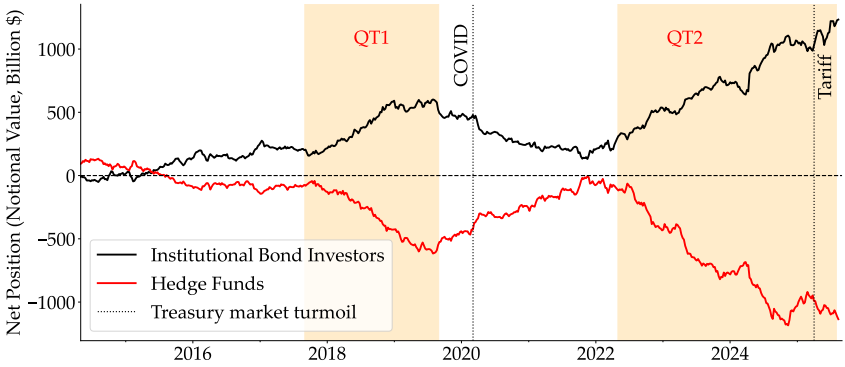
But the recent QT episode has been anything but quiet.

Fact 1. Inelastic Demand in the Treasury Market



Fact 2: Institutional Bond Investors Increasingly Purchase Treasury Derivatives Rather Than Cash Bonds

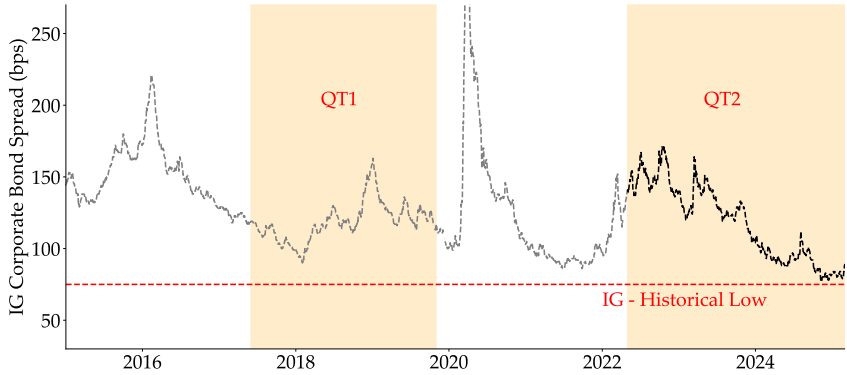
- Fact 1: Inelastic Treasury demand.



◂ Future Trade and Swap Trade ◂ Price Convergence ◂ Treasury Market Turmoil

Fact 3: Credit Spreads Have Fallen to Historical Lows Despite the Fed Tightening

- Fact 1: Inelastic Treasury demand.
- Fact 2: Treasury derivative usage.



◂ Credit Spread Converges to CDS ◂ Treasury CDS

QT: Sharp Rise in Treasury Bonds Relative to Corporate Bonds

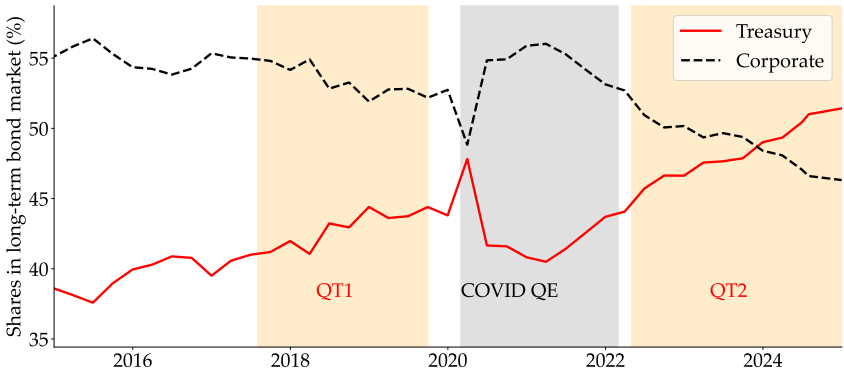


Figure: The Treasury share and corporate bond share in the long-term bond market

Level Supply vs Demand | Bond Market Composition | Corporate Market Timing | Share after 09 crisis

This Paper: Higher Treasury Share, More Segmented Markets

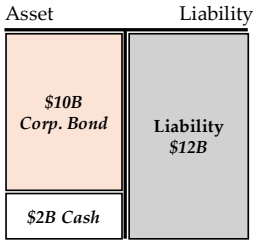
- Institutional bond investors face **duration mandates** and **leverage constraints**.
- Corporate bonds efficiently satisfy duration mandate, also providing credit exposure.
- Corporate bond share $\downarrow \Rightarrow$ investors rely more on Treasuries to meet duration mandates $\Rightarrow$ tighter leverage constraints.

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- Corporate bonds efficiently satisfy duration mandate, also providing credit exposure.
- Corporate bond share $\downarrow \Rightarrow$ investors rely more on Treasuries to meet duration mandates $\Rightarrow$ tighter leverage constraints.
- $\Rightarrow$ weaker substitutability between Treasuries and corporate bonds $\Rightarrow$ more segmented markets
  1. Treasury supply shocks spill over less to corporate bonds and stay in Treasury market.
  2. Constrained investors actively use Treasury derivatives for leverage.

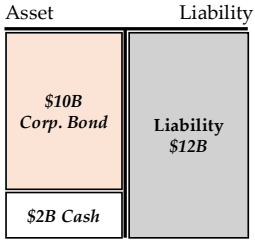
Key Idea: Corporate Bonds as a Preferred "Risk Bundle"

Region 1: high corporate share

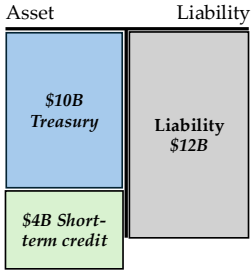


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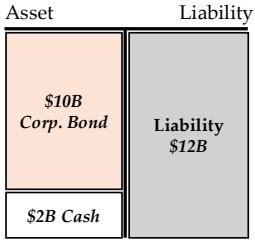
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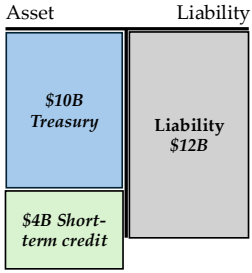
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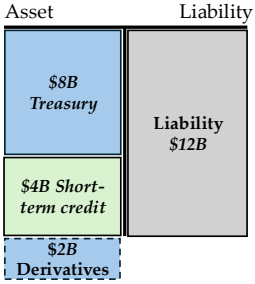


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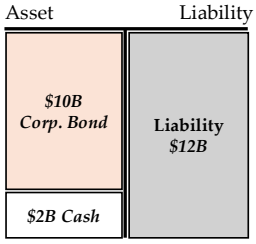
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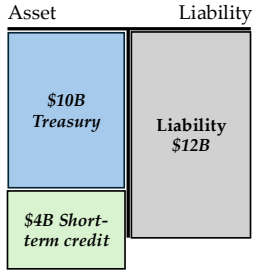
"Synthetic corp. bond" with derivatives

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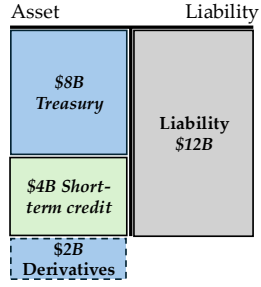


Region 2: low corporate share



"Synthetic corp. bond"

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"Synthetic corp. bond" with derivatives

Region 2: rebalance less between Treasuries and corporate bonds.

Duration Mandate and Derivative Usage in Bond Investors' Words

The 2024 US Treasury Market Conference

"...we are looking to achieve a portfolio that has the same cash flow dynamics and interest rate dynamics as our liabilities...We are going to incorporate spread assets...into that asset allocation, and a lot of those [assets] will be shorter duration or intermediate duration...We look to the derivatives markets to adjust the duration of our portfolios."

—— Joseph Demetrick, senior vice president at MetLife

"...asset managers measure their performance against a benchmark that includes a high percentage of medium-term and long-term Treasuries...They invest in corporate bonds and other fixed income securities with credit risk because they offer higher yields than Treasuries. ...they use Treasury futures to replicate the Treasury exposure. That allows them to keep the credit exposure overweight that generates higher returns but still maintain the target level of duration...

—— David Rogal, head of total return and inflation portfolios at BlackRock

◂ MF Allocation in Data

Results Preview

Model: The composition of asset supply shapes the degree of market segmentation

1. State-dependent Treasury supply effects across:
   - (1) Treasury market, (2) corporate bond market, (3) Treasury derivative market.
2. Hedge fund basis trade.

Empirical Evidence:

1. Aggregate level:
   - Novel Treasury supply shocks constructed from the Treasury issuance calendar.
   - Stronger effects on Treasury yield, but weaker pass-through to corporate bond market when bond investors are more constrained.
2. Micro-level evidence from data on bond investors' derivative positions.

Policy Implications: QE–QT asymmetry and optimal QT timing.

◂ References ◂ Closest Related Literature

Theoretical Framework

Institutional Background

1. Fixed-income markets are *partially segmented* across asset classes and maturities.

2. Mutual funds, pension funds, and insurance companies trade across markets:
   - hold more than 60% of long-term Treasuries and corporate bonds;
   - are subject to **duration mandates** and **leverage constraints**.

	Duration mandate	**(Financial) leverage constraint**
Bond Mutual Funds	Bond benchmark index	Investment Company Act of 1940
Pension Funds	Retirement benefit payments	Employee Retirement Income Security Act of 1974
Insurance Companies	Life insurance policies and annuities	NAIC risk-based capital regulation; State-level insurance laws on borrowing limits

Their aggregate debt-to-asset ratio is below 1.5% (Financial Stability Board, 2023).

◂ Bond Allocators Share ◂ Facts from Micro Data

Model Setup: Bond Supply

Three periods $t \in \{0, 1, 2\}$. At $t = 0$, the interest rate is fixed at i_0.

1. **Government (and the Fed)** supplies
   - long-term Treasury bonds (T) with quantity S_T,
   - short-term Treasury bills, with supply elastic to market demand.
2. **Representative firm** fulfills its financing needs by issuing
   - short-term corporate bonds (S) with quantity S_S,
   - long-term corporate bonds (L) with quantity S_L.

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   - long-term corporate bonds (L) with quantity S_L.

- **Credit risk**: the firm may default at $t = 1$, and the log default loss follows

$$z_1 \sim \mathcal{N}(\bar{z}, \sigma_z^2), \qquad \bar{z} > 0.$$

- **Duration risk**: the interest rate at $t = 1$ follows $i_1 = i_0 + \epsilon_1^i$,

$$\epsilon_1^i \sim \mathcal{N}(0, \sigma_i^2).$$

Model Setup: Bond Supply

- Three risky assets:
  - Long-term Treasury: duration risk.
  - Long-term corporate bond: duration risk and credit risk $\Rightarrow$ **"risk bundle"**.
  - Short-term corporate bond: credit risk.

- All bonds are zero-coupon bonds with face value equal to 1.

- One-period log excess returns follow

$$
\begin{aligned}
ex_1^T &= 2y_0^T - i_1 - i_0 \\
ex_1^L &= 2y_0^L - i_1 - z_1 - i_0 \\
ex_1^S &= y_0^S - z_1 - i_0
\end{aligned}
$$

Model Setup: Bond Demand

Two kinds of investors:

1 **Bond allocator** can invest across markets (arbitrageur à la Vayanos and Vila (2021)).

$$\max_{w_T, w_L, w_S} \mathbb{E}_0[ex_1^{\text{portfolio}}] - \frac{\gamma}{2} Var_0[ex_1^{\text{portfolio}}]$$

$$\text{s.t.} \quad ex_1^{\text{portfolio}} = w_T ex_1^T + w_L ex_1^L + w_S ex_1^S$$

$$w_T + w_L = \bar{D} \qquad \textbf{[Duration Mandate]}$$

$$w_T + w_L + w_S \leq 1 \qquad \textbf{[Leverage Constraint]}$$

- *Cross-market bond mutual funds*
- *Insurance companies and pension funds*

◂ Micro foundation: duration mandate

Model Setup: Bond Demand

Two kinds of investors:

1. **Bond allocator** can invest across markets (arbitrageur à la Vayanos and Vila (2021)).
2. **Specialized investor** g in market $g \in \{T, L, S\}$:

$$D_g = \varepsilon_g \mathbb{E}_0[ex_1^T], \quad \forall g \in \{T, L, S\}.$$

- *Foreign reserve managers* in government bonds.
- *Mandate-driven investors* (Gabaix and Koijen, 2021).

Equilibrium Results under Two Regions

Assumption (Fixed short-term credit spread)

Short-term corporate bond specialists have perfectly elastic demand: $\varepsilon_S \to \infty$. Hence fix the short-term credit spread at a constant level.

Proposition (Equilibrium under two regions)

*The economy features a **slack region** and a **binding region**. Under the **binding region**, bond allocators hit the leverage constraint, and Treasury supply shocks have:*

i) *a **stronger** price impact on Treasury yield.*

ii) *a **weaker** price impact on long-term corporate bond yield.*

◂ Elasticity in math

Leverage Constraint Binds when Treasury Share is High

Proposition (Boundary condition for two regions)

Denote $\bar{S}_{Long} \equiv S_L + S_T$. The boundary condition between slack region and binding region follows

$$g(\bar{S}_{Long}, S_T, \bar{D}) \equiv \underbrace{(\varepsilon_L + \varepsilon_T) S_T}_{\text{Treasury share}} + \underbrace{\varepsilon_T (\bar{D} - \bar{S}_{Long})}_{\text{Duration demand relative to supply}} + const. \begin{cases} > 0 & \text{(binding region)}, \\ < 0 & \text{(slack region)}. \end{cases}$$

More likely entering binding region under:

1. High Treasury share $\Rightarrow$ use more Treasuries for duration mandate.
2. High duration mandate ($\bar{D}$) relative to total duration supply ($\bar{S}_{Long}$).

◂ Comparative analysis

Allocator Portfolio Balancing Relates Three Markets

- Allocator ensures "no-arbitrage" between long- and short-term bonds:

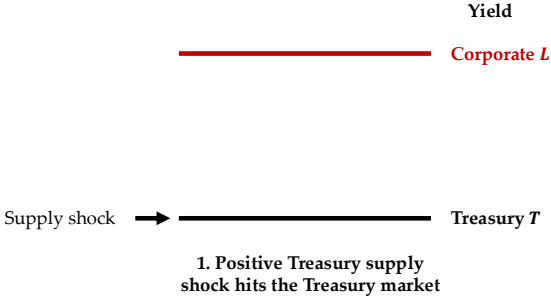
$$\underbrace{\mathbb{E}_0[ex_1^L] - \mathbb{E}_0[ex_1^T]}_{\text{long-term credit spread}} = \underbrace{\mathbb{E}_0[ex_1^S]}_{\text{short-term credit spread}} - \underbrace{\lambda}_{\substack{\text{shadow cost of} \\ \text{leverage constraint}}}$$

  - $\lambda > 0$: accept lower spread for the long-term credit $\rightarrow$ preference for long-term corp.

◂ Data: term structure of credit spread

How do Treasury Supply Shocks Affect Bond Yields?

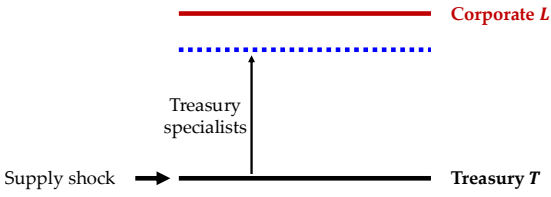
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1. Positive Treasury supply shock hits the Treasury market

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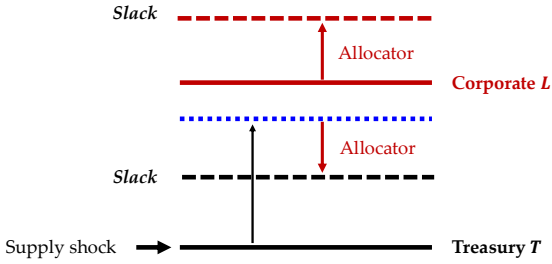
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2. Undiversified treasury specialists drive up treasury yield

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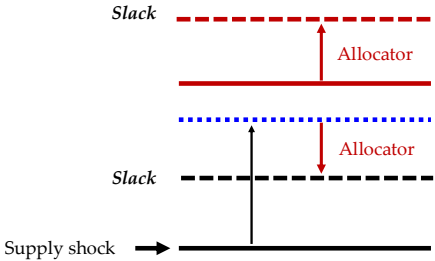
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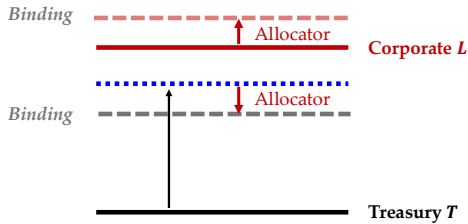
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4. Binding region: less rebalancing

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5. Supply shock in the binding region: (1) greater price impacts on Treasuries and (2) narrower long-term credit spread.

Extension: Model with Treasury Derivatives

Hedge funds (and dealers): long cash Treasuries (T) and short Treasury derivatives (D)

- Choose the derivative supply S_D:

$$\max_{S_D} \quad S_D \mathbb{E}_0[ex_1^T - ex_1^D] - \frac{\kappa}{2}(S_D\sigma)^2$$

  - $\frac{\kappa}{2}(S_D\sigma)^2$: balance sheet costs of holding cash bond/risk of basis trade.

◂ Intermediation cost

- Derivative supply curve:

$$S_D = \varepsilon_D \mathbb{E}_0[ex_1^T - ex_1^D], \text{ with supply elasticity } \varepsilon_D \equiv \frac{1}{\kappa\sigma^2}$$

◂ Price convergence ◂ Future basis trade and swap trade

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Allocator: duration mandate can be met with cash bonds or derivatives:

$$\textbf{[Duration Mandate]} \qquad w_L + w_T + w_D = \bar{D}$$

Benefit of Basis Trade: Partially Restore Treasury Market Elasticity

Proposition (Basis trade **partially** restores Treasury market elasticity)

$$\textit{Treasury market elasticity}: \underbrace{\varepsilon_{TT}^{Slack}}_{\text{slack region}} > \underbrace{\varepsilon_{TT}^{BasisTrade}}_{\text{binding region + basis trade}} > \underbrace{\varepsilon_{TT}^{Binding}}_{\text{binding region}}$$

Bond market "No-arbitrage condition":

$$\underbrace{\mathbb{E}_0[ex_1^L] - \mathbb{E}_0[ex_1^T]}_{\text{long-term credit spread}} = \underbrace{\mathbb{E}_0[ex_1^S]}_{\text{short-term credit spread}} - \underbrace{\mathbb{E}_0[ex_1^T] - \mathbb{E}_0[ex_1^D]}_{\text{derivative spread}}$$

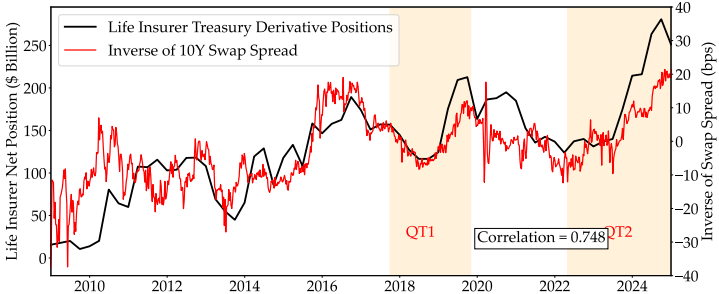
- Derivative spread = rental cost of hedge funds/dealers' balance sheets.

◂ Elasticity in math

Derivative Spread as Allocators' Leverage-constraint Measure

Proposition (Basis Trade and Derivative Spread)

1. *Higher Treasury share and duration mandate $\bar{D} \Rightarrow$ more constrained balance sheet $\Rightarrow$ higher derivative positions.*
2. *Higher derivative positions $\Rightarrow$ wider derivative spread charged by HFs/Dealers.*



Summary: Model Predictions to Test with Data

1. **Bond allocators become more leverage constrained when:**
   - the Treasury share in the debt market S_T is large.
   - facing a higher duration mandate $\bar{D}$.

2. **When they are more constrained:**
   - take larger positions in Treasury derivatives, fueling basis trade;
   - $\Rightarrow$ Treasury derivative spreads widen $\Rightarrow$ proxy for constraints *in the aggregate level*.

◂ Robustness: future spread, demand-supply decomposition.

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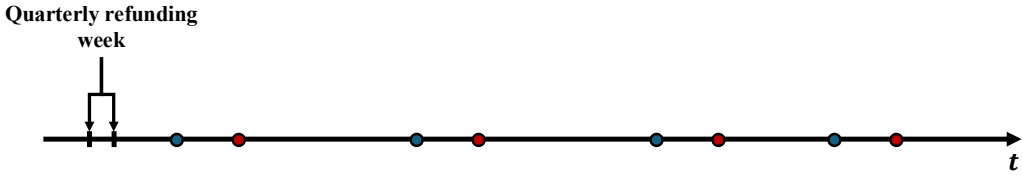
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◂ Robustness: future spread, demand-supply decomposition.

3. **Consequences of tighter allocator constraints:** Treasury supply shocks
   - have a larger price impact on Treasuries;
   - have a smaller price impact on corporate bonds, compressing credit spreads.

Empirical Evidence

Treasury Supply Shocks from Quarterly Treasury Refunding Cycle



Step 1: Quarterly refunding week ⇒ Supply shocks 1

1. Monday: borrowing estimate for the current quarter and next quarter.

$$\text{Shock } 1_t = \frac{\widetilde{\text{Borr. Est}_t}}{D_t} = \frac{\text{Borr. Est}_t^t - \text{Borr. Est}_{t-1}^t}{D_t}$$

2. Wednesday: allocation of borrowing estimate into different bonds.

Treasury Supply Shocks from Quarterly Treasury Refunding Cycle

February 2, 2004
JS-1133

Treasury Announces Market Financing Estimates

The Treasury Department announced today that it expects net borrowing of marketable debt to total $177 billion in the January – March 2004 quarter. The projected cash balance on March 31 is $20 billion. In the last quarterly announcement on November 3, 2003, Treasury announced that it expected net borrowing to total $160 billion with an end-of-quarter cash balance of $20 billion. This increase in borrowing is due to lower receipts, primarily from an increase in tax refunds, and higher outlays.

Treasury also announced that it expects net borrowing of marketable debt to total $75 billion in the April – June 2004 quarter. The projected cash balance on June 30 is $45 billion.

During the October – December 2003 quarter, Treasury's net marketable borrowing totaled $113 billion and the cash balance on December 31 was $33 billion. On November 3, Treasury announced that it expected net marketable borrowing to total $117 billion with an ending quarter cash balance of $35 billion. The decrease in borrowing is primarily attributable to lower outlays.

Additional financing details relating to Treasury's Quarterly Refunding will be released at 9:00 A.M. on Wednesday, February 4.

US TREASURY FINANCING SCHEDULE FOR 1ST QUARTER 2004
BILLIONS OF DOLLARS

ISSUE	ANNOUNCEMENT DATE	AUCTION DATE	SETTLEMENT DATE	OFFERED AMOUNT 4-WK	3-MO	6-MO	MATURING AMOUNT	NEW MONEY
4-WEEK AND	12/24	12/29	1/2	12.00 A	16.00 A	15.00 A	51.00	-8.00
3&6 MONTH BILLS	12/31	1/5	1/8	9.00 A	17.00 A	16.00 A	45.00	-3.00
	1/8	1/12	1/15	8.00 A	17.00 A	16.00 A	44.00	-3.00
	1/15	1/20	1/22	9.00 A	18.00 A	16.00 A	46.00	-3.00
	1/22	1/26	1/29	14.00 A	18.00 A	16.00 A	46.00	2.00
	1/29	2/2	2/5	19.00 A	19.00 A	17.00 A	43.00	12.00
	2/5	2/9	2/12	20.00	19.00	18.00	40.00	17.00
	2/12	2/16	2/19	22.00	19.00	20.00	41.00	20.00
	2/19	2/23	2/26	24.00	19.00	20.00	46.00	17.00
	2/26	3/1	3/4	25.00	19.00	20.00	51.00	13.00
	3/4	3/8	3/11	25.00	19.00	20.00	51.00	13.00
	3/11	3/15	3/18	22.00	18.00	18.00	51.00	7.00
	3/18	3/22	3/25	18.00	18.00	18.00	54.00	0.00
					693.00		609.00	84.00
CASH MANAGEMENT BILLS								
13-Day Bill	12/29 Matures 1/15	12/30	1/2		15.00		15.00	0.00
11-Day Bill	3/2 Matures 3/15	3/3	3/4		20.00		20.00	0.00
COUPONS						CHANGE IN SIZE		
5-Year Note	1/5	1/7	1/15		16.00 A		15.22	0.78
10-Year TIPS	1/5	1/8	1/15		12.00 A	+ 1.00		12.00
2-Year Note	1/26	1/29	2/2		26.00 A		24.55	1.45
3-Year Note	2/4	2/10	2/17		24.00			24.00
5-Year Note	2/4	2/11	2/17		16.00			16.00
10-Year Note	2/4	2/12	2/17		17.00		11.82	5.18
2-Year Note	2/23	2/25	3/1		26.00		23.74	2.26
5-Year Note	3/8	3/15	3/17		16.00			16.00
10-Year Note (R)	3/8	3/15	3/22		13.00			13.00
2-Year Note	3/22	3/24	3/31		27.00	1.00	24.54	2.46
					193.00		99.87	93.13

R = Reopening
A = Announced

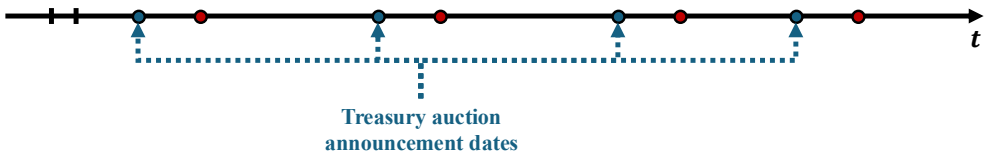
Treasury announced a Q1 borrowing need of $177 billion on 2/2/04

NET CASH RAISED THIS QUARTER: 177.13

Refunding week in 2004 Q1:

$$\widetilde{\text{Borr. Est}}_t = \text{Borr. Est}_t^t - \text{Borr. Est}_{t-1}^t = \$177 \text{ B} - \$160 \text{ B} = \$17 \text{ B}$$

Treasury Supply Shocks from Quarterly Treasury Refunding Cycle



Step 2: Auction announcement ⇒ Supply shocks 2

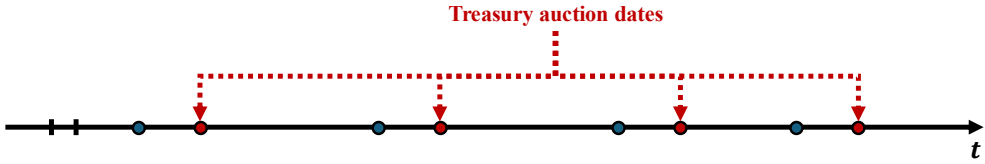
- announce **the real offering amount** several days before auction;
- announce 1–3 bonds simultaneously

$$\text{Shock } 2_t = \sum_m \frac{Duration_m}{Duration_{10Y}} \tilde{S}_{m,t} = \sum_m \frac{Duration_m}{Duration_{10Y}} \left(S_{m,t} - S_{m,t}^{\text{tentative}} \right)$$

Example: Auction Ann Auction Results

Time Series: Refunding Ann Auction Ann

Treasury Supply Shocks from Quarterly Treasury Refunding Cycle



Step 3: Auction result release ⇒ Supply shocks 3

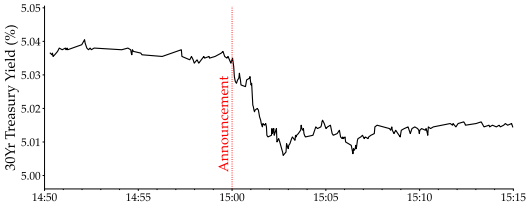
- release the winning yield and **bid-to-cover ratio** within 2 mins after the auction close.
- **bid-to-cover ratio** summarizes auction results (Ray et al., 2024).

Example: Auction Ann Auction Results

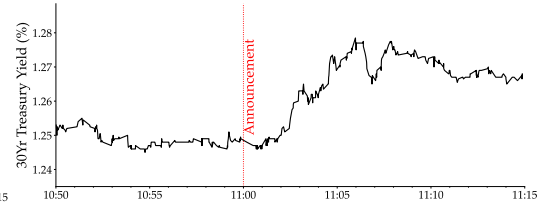
Time Series: Refunding Ann Auction Ann

Intraday Treasury Price Response to Shocks

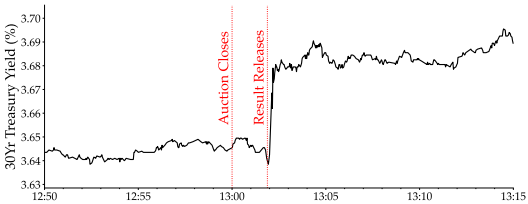
Supply shocks 1 — Oct 30, 2023:
$76bn below the prior-quarter forecast.



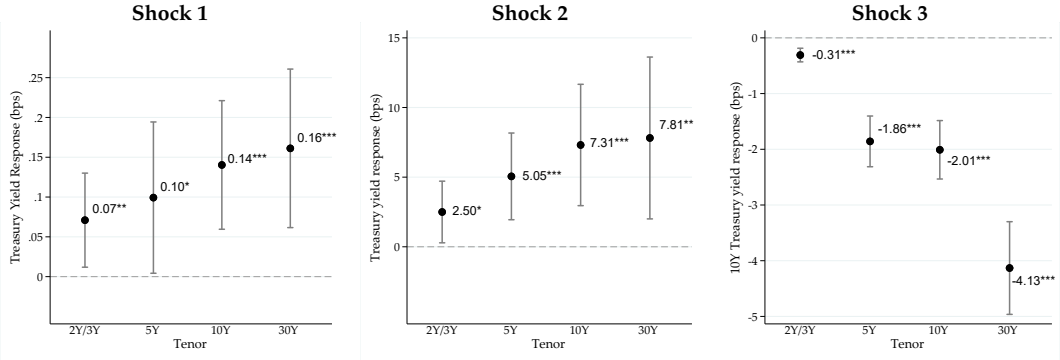
Supply shocks 2 — Apr 2, 2020:
+$1bn 10-year and +$1bn 30-year vs. schedule.



Supply shock 3 — 30-year auction, Feb 9, 2023:
bid-to-cover in the bottom 25% historically.



Average High-frequency Effects by Tenor



In terms of 10Y-equivalent Treasuries

- Shock 1: Supply ↑ by \$100B ⇒ 10Y yield ↑ 0.53 bps (**lower bound**).
- Shock 2: Supply ↑ by \$100B ⇒ 10Y yield ↑ 7.31 bps.
- Shock 3: Bid submissions ↓ by \$100B ⇒ 10Y yield ↑ 8.33 bps.

◂ Supply effects in the literature ◂ Robustness: alternative measures ◂ Conversion

Less Elastic Treasury Market when Allocators are more Constrained

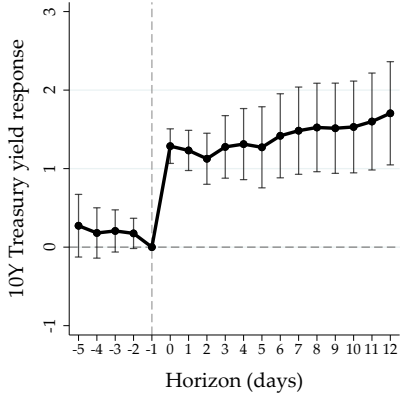
- Most constrained: $Q_{swap\ spread} = 1$.
- Least constrained: $Q_{swap\ spread} = 4$.
- Auction days: supply shocks under the most constrained region are 1.7× stronger than under the least constrained region.

	(1) Shock 1	(2) Shock 2	(3) Shock 3
VARIABLES	$\Delta y_{10\ min}$	$\Delta y_{10\ min}$	$\Delta y_{10\ min}$
$Shock_t$	0.677**	30.04**	-2.522**
	(0.266)	(11.98)	(0.518)
$Shock_t \times Q_{Swap\ Spread_t^{10Y}} = 2$	-0.536*	-24.77**	0.730*
	(0.289)	(12.17)	(0.314)
$Shock_t \times Q_{Swap\ Spread_t^{10Y}} = 3$	-0.555**	-25.27*	1.264***
	(0.272)	(13.61)	(0.317)
$Shock_t \times Q_{Swap\ Spread_t^{10Y}} = 4$	-0.859***	-29.74**	1.020**
	(0.292)	(13.45)	(0.314)
$Q_{Swap\ Spread_t^{10Y}} = 2$	0.362	-0.206	-1.742*
	(0.229)	(0.330)	(0.890)
$Q_{Swap\ Spread_t^{10Y}} = 3$	0.428**	-0.178	-2.724**
	(0.178)	(0.340)	(0.924)
$Q_{Swap\ Spread_t^{10Y}} = 4$	0.281	-0.237	-2.107*
	(0.178)	(0.352)	(0.953)
Observations	366	725	1,096
R-squared	0.118	0.112	0.116
Maturity FE	Y	Y	Y

◂ Margin plots ◂ Alternative specification: Treasury share

Treasury Auction Supply Effects are Persistent

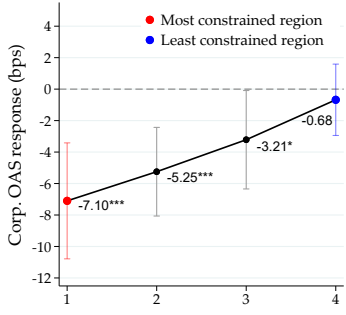
$$\Delta y^{10Y}_{t-1\to t+h} = \alpha_h + \beta_h \Delta y^{10Y,10\text{min}}_{t} + \text{Controls}_t + \varepsilon_{t,h}$$



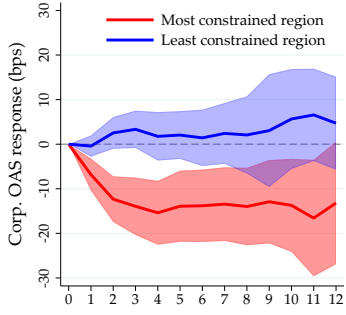
Weaker Pass-through to Corporate Bonds Under Constrained Region

$$\text{Auction day-}t: \quad \Delta OAS_{t-1\to t+h} = \alpha_h + \beta_h \Delta y_t^{10Y} + \sum_{j=1}^{3} \Delta OAS_{t-j-1\to t-j} + \text{Controls}_t + \varepsilon_{t,h}$$

(a) Auction Day t



(b) Impulse Response Functions

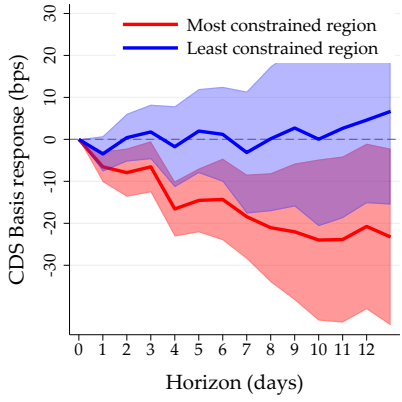


Most constrained region: 1% ↑ in 10Y Treasury yield on auction days,

- AAA bond credit spread ↓ 7.10 bps in the same day, ↓ 12 bps in a week

◂ High frequency shock ◂ Lower Rating Bonds ◂ Long-term Bond Auction

Weaker Pass-through to Corporate Bonds Under Constrained Region



Remove the credit default component:
IG Bond CDS Basis = Credit Spread - IG Bond CDX spread.

◂ Time series: Credit Spread Converges to CDS

Does Micro Data Support the Mechanisms?

1. Global Bond Mutual Funds Holdings from MorningStar.
   - Monthly/quarterly asset holdings starting from 2008Q1-2024Q4.
   - $\approx$ 60% of funds provide details of derivative holdings.
     - Derivative positions: contract name, direction, notional amount and market value.
     - Identify 156,664 long Treasury future positions and 103,588 short positions by name matching.

2. US insurance company holdings from NAIC statutory filings.
   - NAIC schedule D: Quarterly asset holdings starting from 2008Q1-2024Q4.
   - NAIC schedule DB: detailed derivatives positions on future and interest rate swaps.
     - Identify 32,497 long Treasury futures positions and 19,002 short positions.
     - Identify 505,485 received-fixed IRS positions and 384,781 pay-fixed swap positions.

Investors' Bond and Derivative Positions Align with Model Predictions

Bond Mutual Funds:

- Future users are primarily actively-managed funds benchmarked to Bloomberg aggregate bond index, with duration between 6–8 year. ◂ Derivative Users
- Underweight Treasuries and overweight corporate bonds for outperforming benchmark index. ◂ Active Allocation
- Face duration gap when corporate bonds are scarce, actively use Treasury futures for duration exposure. ◂ Mutual fund future usage ◂ Future Positions by Tenor

Life Insurance companies:

- Target at long-term liability (typically 15—25 years.) ◂ Duration distribution
- Primarily use interest rate swaps for duration exposures. ◂ Insurer derivative usage

◂ Return to Institutional Background

More Derivative Usage when Treasuries Crowd the Debt Market

	Mutual Funds				Life Insurance Companies			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
VARIABLES	Derivative dollar duration (DV01 in k$)				Derivative dollar duration (DV01 in k$)			
$Swap\ Spread_t^{10Y}$	-1.516***			-0.915**	-26.94**			-22.09***
	(0.413)			(0.454)	(11.56)			(4.726)
$\Delta\ Treasury\ Share_t$		37.75***		20.83*		464.2***		203.8**
		(10.71)		(11.82)		(93.74)		(91.14)
$Bond\ index\ duration_t$			36.14***	23.07**				
			(12.93)	(10.14)				
y_t^{10Y}							-274.0***	-307.4***
							(86.71)	(69.57)
Observations	46,197	46,197	46,197	46,197	7,107	7,107	7,107	7,107
R-squared	0.401	0.401	0.401	0.402	0.514	0.514	0.513	0.516
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y

Policy Implications

Model Implies a QE-QT Asymmetry

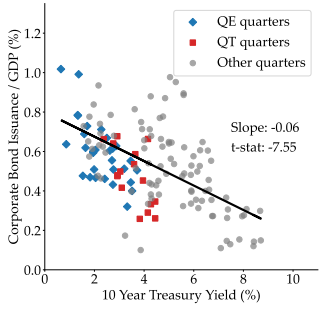


Figure: Quarterly Long-term Corporate Bond Issuance (>10Y).

- QE: low interest rate ⇒ high corporate bond supply ⇒ increasing corporate share.
- QT: high interest rate ⇒ low corporate bond supply ⇒ increasing Treasury share.

Model Implies a QE-QT Asymmetry

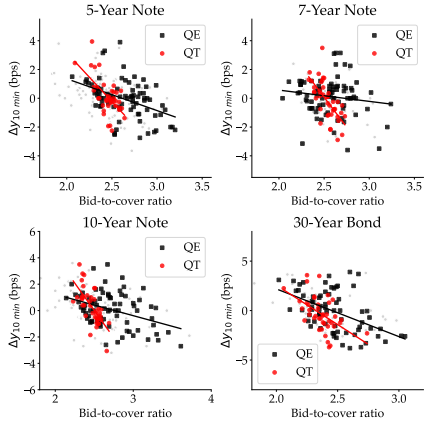
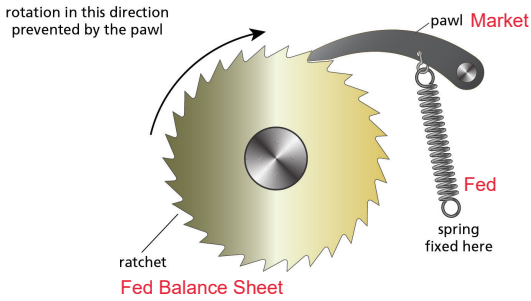


Figure: Treasury Market is Less Elastic under QT than QE.

- 10Y Treasury bid ↓ by \$100B ⇒ 10Y yield ↑ **25.7 bps (QT)** vs ↑ **11.1 bps (QE)**

Model Implies a QE-QT Asymmetry



"[QE] tends to be deployed in response to bad news, but isn't reversed when the bad news ends. As a result, the stock of bonds held by central banks ratchets up, expanding their balance sheets into the longer term."

– Mervyn King (former Governor, Bank of England), *Quantitative Easing Is a Dangerous Addiction*, 2021

How Can Policymakers Manage Basis Trade Risks?

Should we fully restrict basis trade?

No, basis trade reduces Treasury refunding cost.

Timing	Policy	Channel
Ex ante	Timing QT: offloading before rate hikes	Offloading Treasuries when corporate bonds are not scarce.
	1. Develop a central-clearing repo market 2. Exempt dealers' Treasury positions in trading accounts from SLR (Liang and Zhu, 2025)	Improve dealers' Treasury absorption capacity when basis trade unwinds.
Ex post	Backstop HF basis trade positions (Kashyap et al, 2025)	Purchase cash bonds **&** sell derivative simultaneously.

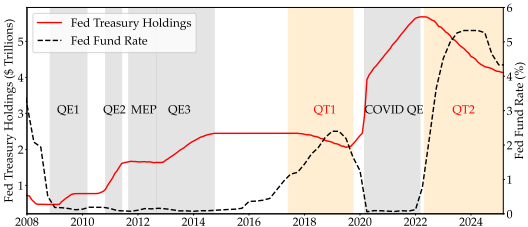
Conclusion: When Treasuries Dominate the Debt Market

- High Treasury share raises debt market segmentation.
- Treasury supply shocks have big price impacts on Treasuries and weak pass-through to corporate bonds.
- Institutional bond investors leverage via Treasury derivatives, fueling basis trade.
- QE & QT effects depend on debt market structure.

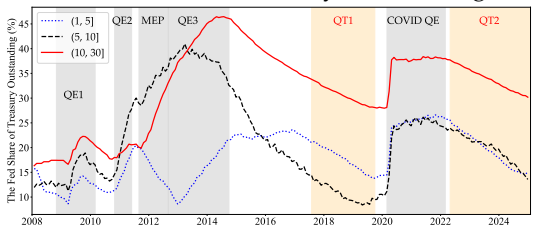
Appendix

Treasury holdings of the Federal Reserve

(a) Treasury holdings amount

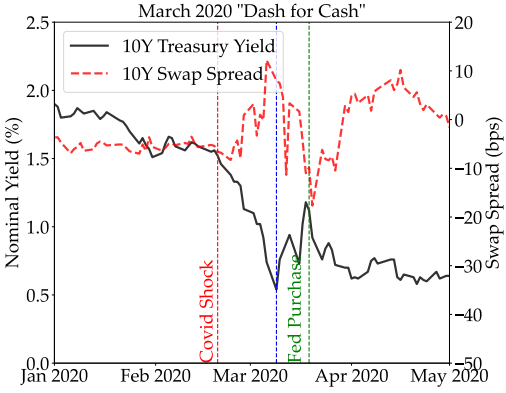


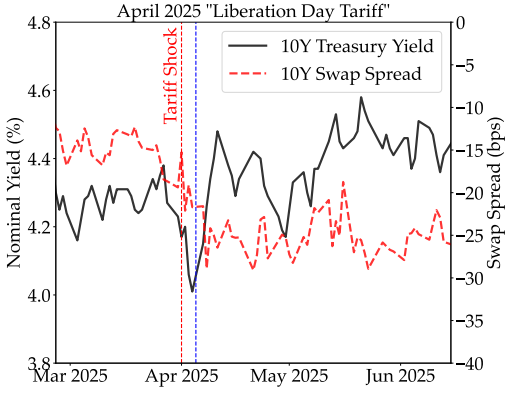
(b) Share in the Treasury outstanding



Return

US Treasury market Turmoil in March 2020 and April 2025





Return

Basis Trade

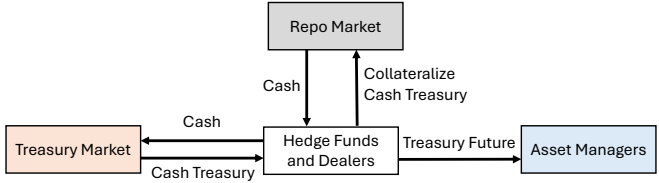


Figure: Treasury future

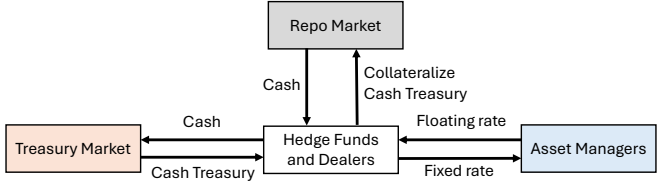


Figure: Interest rate swap

Return to Opening Return to Model

Convergence Trade on Treasury Futures

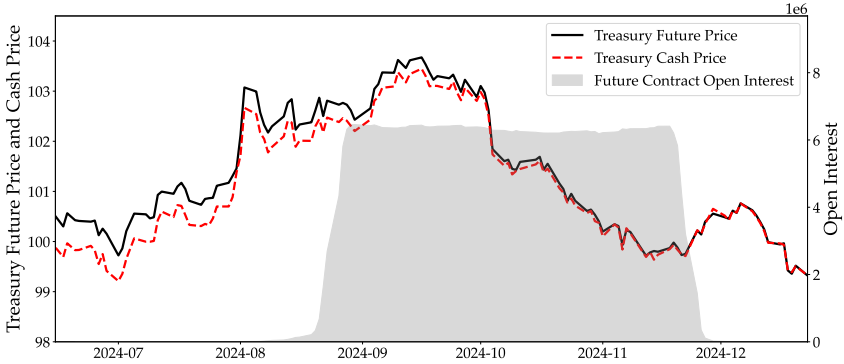


Figure: Convergence of Treasury Future Prices and Cash Price

Return Return to Model

Dealers and Hedge Funds' Intermediation Costs

1. Dealers: **supplementary leverage ratio (SLR)** requirement by holding cash Treasuries on the balance sheet.

2. Hedge funds are subject to (Market Risk Advisory Committee, 2024):

   - **Counterparty risks.**
   - **Repo rollover risks.**
   - **Margin requirements.**

Return

The Long-term IG Credit Spread Converges to the CDS Spread

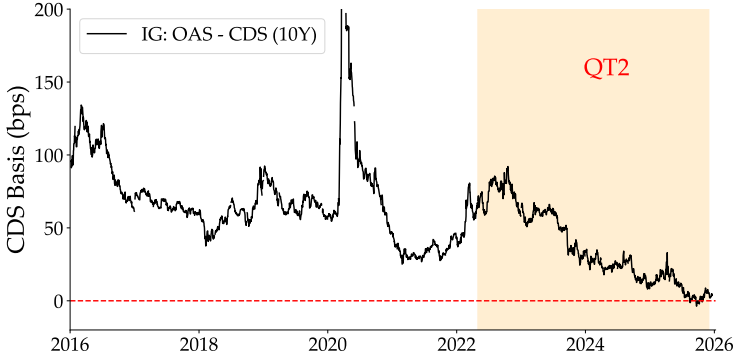
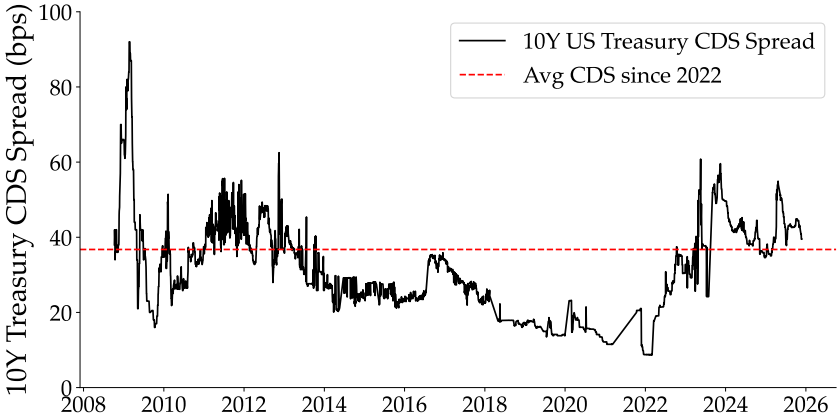


Figure: 10–15Y US IG corporate credit spread minus 10Y IG Bond CDX spread.

- Credit spread typically exceeds CDS spread, reflecting funding and liquidity premia associated with holding cash bonds.

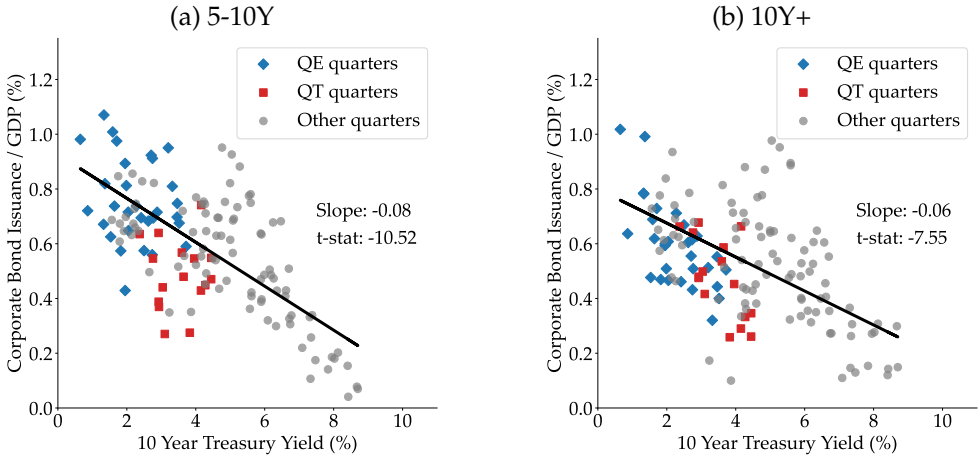
Return to opening Return to passthrough

10Y Treasury Credit Default Spread



Return

Corporate Market Timing: Corporate Bond Quarterly Supply



Data source: LSEG Debt New Issues Deals Data. 1990Q1-2024Q4.

Return

Composition of US Bond Market

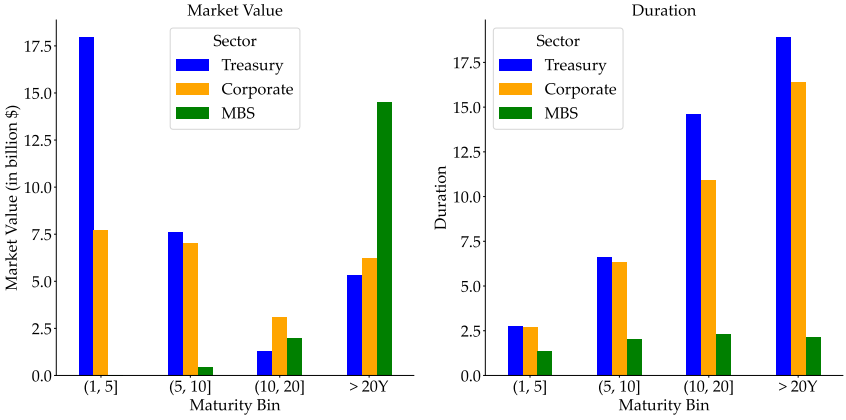


Figure: US bond market composition (2021Q1)

Note: The figure shows the composition of the iShares Core US Aggregate Bond ETF.

Return

References

Market segmentation theory

- Preferred-habitat: Tobin (1958, 1969); Vayanos and Vila (2021); Greenwood et al. (2023, 2018); Ray et al. (2024)
- Arbitrageur balance sheet friction: He et al. (2022); Kekre and Lenel (2022)

Unconventional monetary policy

- Effects: Gagnon et al. (2011); Vissing-Jorgensen and Krishnamurthy (2011); Swanson (2011); D'Amico and King (2013); Du et al. (2024); D'Amico and Seida (2024)...
- Pass-through: Gilchrist and Zakrajšek (2013); Gilchrist et al. (2015); Acharya et al. (2025); Eren et al. (2025)
- State-dependence: Acharya and Rajan (2024); Haddad et al. (2024); Jiang and Sun (2024)

Treasury supply effects

- Treasury supply shocks: Greenwood et al. (2010); Wang and Zhao (2024); Lou et al. (2013); Fleming and Liu (2016); Ray et al. (2024); Phillot (2025); Wiegand (2025)
- Crowding in and crowd out: Friedman (1978); Pinardon-Touati (2021); Li et al. (2025)

Interest rate derivative basis trade

- Hedge funds: Schrimpf et al. (2020); Barth and Kahn (2020); Vissing-Jorgensen (2021)
- Policy implications: Kashyap et al. (2025); Liang (2025)

References and Positioning

Market segmentation

- Greenwood et al. (2018); Ray et al. (2024)
- This paper: the composition of asset supply endogenously shapes the degree of market segmentation.

Unconventional monetary policy

- Acharya and Rajan (2024); Haddad et al. (2024); Jiang and Sun (2024)
- This paper: QE–QT asymmetry arises from debt-market structure and investor constraints.

Treasury supply effects

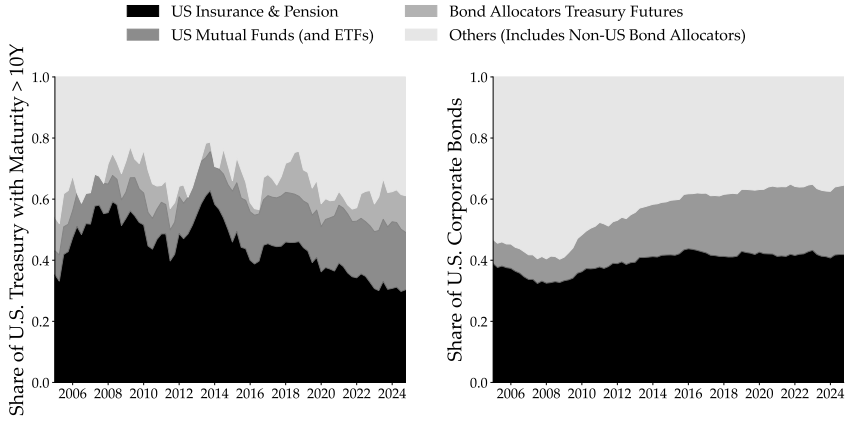
- Wang and Zhao (2024); Lou et al. (2013); Ray et al. (2024); Phillot (2025)
- This paper: comprehensive Treasury shock series from issuance calendars & high-frequency identification.
- Corporate bonds' "risk-bundle" property dampens government-debt crowding-out effects.

Interest-rate derivative basis trades

- Barth and Kahn (2020); Kashyap et al. (2025)
- This paper: explains time variation in basis-trade activity, rationalizes its benefits and risks, and introduces new granular data on asset managers' derivative positions.

Return

Bond Allocators in Treasury and Corporate Bond Markets



Data Source: MorningStar, NAIC Statutory filings, US Financial Account, Traders in Financial Futures Report.

Return

Treasury Borrowing Estimate Announcement on February 2, 2004

February 2, 2004
JS-1133

Treasury Announces Market Financing Estimates

The Treasury Department announced today that it expects net borrowing of marketable debt to total $177 billion in the January – March 2004 quarter. The projected cash balance on March 31 is $20 billion. In the last quarterly announcement on November 3, 2003, Treasury announced that it expected net borrowing to total $160 billion with an end-of-quarter cash balance of $20 billion. This increase in borrowing is due to lower receipts, primarily from an increase in tax refunds, and higher outlays.

Treasury also announced that it expects net borrowing of marketable debt to total $75 billion in the April – June 2004 quarter. The projected cash balance on June 30 is $45 billion.

During the October – December 2003 quarter, Treasury's net marketable borrowing totaled $113 billion and the cash balance on December 31 was $33 billion. On November 3, Treasury announced that it expected net marketable borrowing to total $117 billion with an ending quarter cash balance of $35 billion. The decrease in borrowing is primarily attributable to lower outlays.

Additional financing details relating to Treasury's Quarterly Refunding will be released at 9:00 A.M. on Wednesday, February 4.

US TREASURY FINANCING SCHEDULE FOR 1ST QUARTER 2004
BILLIONS OF DOLLARS

ISSUE	ANNOUNCEMENT DATE	AUCTION DATE	SETTLEMENT DATE	OFFERED AMOUNT 4-WK	3-MO	6-MO	MATURING AMOUNT	NEW MONEY
4-WEEK AND	12/24	12/29	1/2	12.00 A	16.00 A	15.00 A	51.00	-8.00
3&6 MONTH BILLS	12/31	1/5	1/8	9.00 A	17.00 A	16.00 A	45.00	-3.00
	1/8	1/12	1/15	8.00 A	17.00 A	16.00 A	44.00	-3.00
	1/15	1/20	1/22	9.00 A	18.00 A	16.00 A	46.00	-3.00
	1/22	1/26	1/29	14.00 A	18.00 A	16.00 A	46.00	2.00
	1/29	2/2	2/5	19.00 A	19.00 A	17.00 A	43.00	12.00
	2/5	2/9	2/12	20.00	19.00	18.00	40.00	17.00
	2/12	2/16	2/19	22.00	19.00	20.00	41.00	20.00
	2/19	2/23	2/26	24.00	19.00	20.00	46.00	17.00
	2/26	3/1	3/4	25.00	19.00	20.00	51.00	13.00
	3/4	3/8	3/11	25.00	19.00	20.00	51.00	13.00
	3/11	3/15	3/18	22.00	18.00	18.00	51.00	7.00
	3/18	3/22	3/25	18.00	18.00	18.00	54.00	0.00
					693.00		609.00	84.00
CASH MANAGEMENT BILLS								
13-Day Bill	12/29 Matures 1/15	12/30	1/2		15.00		15.00	0.00
11-Day Bill	3/2 Matures 3/15	3/3	3/4		20.00		20.00	0.00
COUPONS						CHANGE IN SIZE		
5-Year Note	1/5	1/7	1/15		16.00 A		15.22	0.78
10-Year TIPS	1/5	1/8	1/15		12.00 A	+ 1.00		12.00
2-Year Note	1/26	1/29	2/2		26.00 A		24.55	1.45
3-Year Note	2/4	2/10	2/17		24.00			24.00
5-Year Note	2/4	2/11	2/17		16.00			16.00
10-Year Note	2/4	2/12	2/17		17.00		11.82	5.18
2-Year Note	2/23	2/25	3/1		26.00		23.74	2.26
5-Year Note	3/8	3/15	3/17		16.00			16.00
10-Year Note (R)	3/8	3/15	3/22		13.00			13.00
2-Year Note	3/22	3/24	3/31		27.00	1.00	24.54	2.46
					193.00		99.87	93.13

R = Reopening
A = Announced

Treasury announced a Q1 borrowing need of $177 billion on 2/2/04

NET CASH RAISED THIS QUARTER: 177.13

Refunding week in 2004 Q1:

$$\widetilde{\text{Borr. Est}}_t = \text{Borr. Est}_t^t - \text{Borr. Est}_{t-1}^t = \$177\text{ B} - \$160\text{ B} = \$17\text{ B}$$

Example of Auction Announcement on March 8, 2004

5-YEAR NOTES AND 9-YEAR 11-MONTH 4% NOTES TO BE ISSUED MARCH 15, 2004

March 8, 2004

Offering Amount	$16,000 million	$11,000 million
Maximum Award (35% of Offering Amount)	$ 5,600 million	$ 3,850 million
Maximum Recognized Bid at a Single Yield	$ 5,600 million	$ 3,850 million
NLP Reporting Threshold	$ 5,600 million	$ 3,850 million
NLP Exclusion Amount	None	$ 5,600 million
Description of Offering		
Term and type of security	5-year notes	9-year 11-month 4% notes (reopening)
Series	F-2009	B-2014
CUSIP number	912828 CC 2	912828 CA 6
Auction date	March 10, 2004	March 11, 2004
Issue date	March 15, 2004	March 15, 2004
Dated date	March 15, 2004	February 15, 2004
Maturity date	March 15, 2009	February 15, 2014
Interest rate	Determined based on the highest accepted competitive bid	4%
Amount currently outstanding	Not applicable	$17,082 million
Yield	Determined at auction	Determined at auction
Interest payment dates	September 15 and March 15	August 15 and February 15
Minimum bid amount and multiples	$1,000	$1,000
Accrued interest payable by investor	None	$3.18681 per $1,000 (from February 15 to March 15, 2004)
Premium or discount	Determined at auction	Determined at auction
STRIPS Information:		
Minimum amount required	$1,000	$1,000
Corpus CUSIP number	912820 JZ 1	912820 JX 6
Due date(s) and CUSIP number(s) for additional TINT(s)	March 15, 2009 - - 912833 2B 5	Not applicable

The realized offering amounts for 10YR note was \$11 Billion, 2 billion less than tentative auction plan. The resulting surprises are

$$\tilde{S}_{\text{short-term},t} = \$0 \text{ billion}, \tilde{S}_{\text{short-term},t} = -\$2 \text{ billion}$$

Return

Example of Auction Result Release on March 11, 2004

BUREAU OF THE PUBLIC DEBT - WASHINGTON DC

FOR IMMEDIATE RELEASE
March 11, 2004

CONTACT: Office of Financing
202-691-3550

RESULTS OF TREASURY'S AUCTION OF 9-YR 11-MONTH NOTES

This issue is a reopening of a note originally issued February 17, 2004.

Interest Rate:	4%	Issue Date:	March 15, 2004
Series:	B-2014	Dated Date:	February 15, 2004
CUSIP No:	912828CA6	Maturity Date:	February 15, 2014

High Yield: 3.752% Price: 102.034

All noncompetitive and successful competitive bidders were awarded securities at the high yield. Tenders at the high yield were allotted 34.31%. All tenders at lower yields were accepted in full.

Accrued interest of $ 3.18681 per $1,000 must be paid for the period from February 15, 2004 to March 15, 2004.

AMOUNTS TENDERED AND ACCEPTED (in thousands)

Tender Type	Tendered	Accepted
Competitive	$ 19,852,035	$ 10,956,550
Noncompetitive	43,495	43,495
FIMA (noncompetitive)	0	0
SUBTOTAL	19,895,530	11,000,045 1/
Federal Reserve	0	0
TOTAL	$ 19,895,530	$ 11,000,045

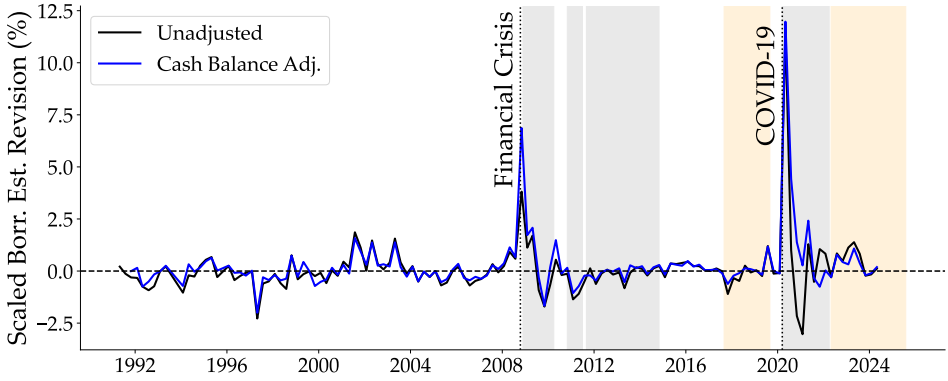
Median yield 3.730%: 50% of the amount of accepted competitive tenders was tendered at or below that rate. Low yield 3.660%: 5% of the amount of accepted competitive tenders was tendered at or below that rate.

Bid-to-Cover Ratio = 19,895,530 / 11,000,045 = 1.81

1/ Awards to TREASURY DIRECT = $26,573,000

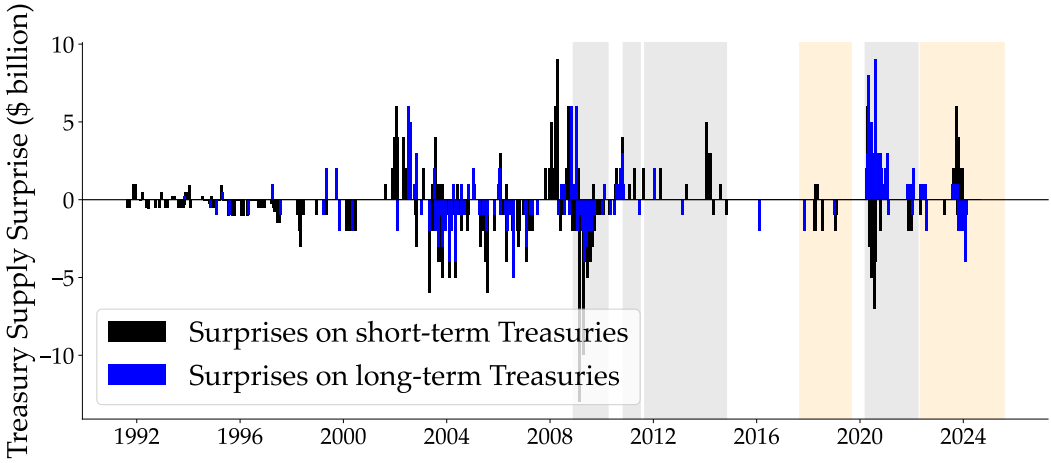
Return

Treasury Supply Shocks on the Quarterly Refunding Dates



Return

Treasury Supply Shocks on the Auction Announcement Dates



Return

Conversion to 10Y-equivalent Treasury

Shock 1: A 1% increase in total borrowing $\Rightarrow$ 10Y yield $\uparrow$ 0.14 bps.

- Average total Treasury debt: $\approx$ \$15T (2008–2024).
- Duration-weighted maturity of quarterly Treasury supply: $\approx$ 1.5 years (2008–2024).
- 1% increase in total borrowing $\approx$ \$26.4B in 10Y-equivalent supply.
- $\Rightarrow$ 10Y-equivalent supply $\uparrow$ by \$100B $\Rightarrow$ 10Y yield $\uparrow$ by $0.14 \times \frac{100}{26.4} = 0.53$ bps.
- Lower bound: assumes issuance shares across maturities are unchanged. In practice, Treasury bills absorb major part of the increase.

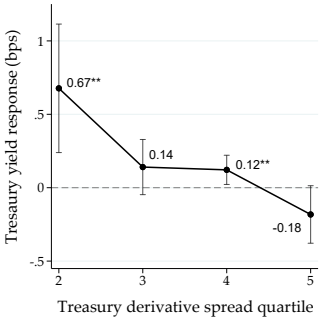
Shock 2: 10Y-equivalent supply $\uparrow$ by \$100B $\Rightarrow$ 10Y yield $\uparrow$ 7.31 bps.

Shock 3: 10Y bid-to-cover $\downarrow$ by 1 $\Rightarrow$ 10Y yield $\uparrow$ 2.0 bps.

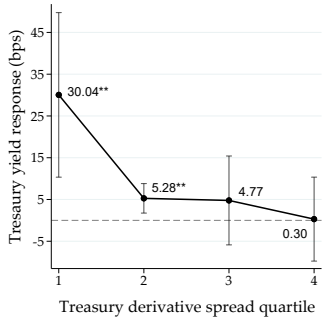
- Average 10Y auction offering: $\approx$ \$24B (2008–2024).
- \$100B decline in bid submissions $\Rightarrow$ 10Y yield $\uparrow$ by $2.0 \times \frac{100}{24} = 8.33$ bps.

Return

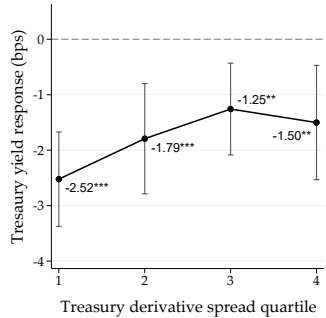
Treasury Market is less elastic when Allocators are more constrained



(a) Shock 1



(b) Shock 2



(c) Auction results in each tenor

Return

QE Literature: Treasury Supply Effect Estimations

QE Operation	Studies	Average Effect Across Studies	Average Effect (per $100bn)
QE1 ($300bn)	47 bps – Gagnon et al. (2011) 41 bps – Krishnamurthy and Vissing-Jørgensen (2011) 35 bps – D'Amico et al. (2012) 30 bps – D'Amico and King (2013) 34 bps – Bonis, Ihrig, and Wei (2017)	37 bps	12.3 bps
QE2 ($600bn)	18 bps – Krishnamurthy and Vissing-Jørgensen (2011) 55 bps – D'Amico et al. (2012) 21 bps – Meaning and Zhu (2011) 15 bps – Swanson (2011) 12 bps – Bonis, Ihrig, and Wei (2017)	24 bps	4 bps
MEP ($667bn)	22 bps – Hamilton and Wu (2012) 17 bps – Meaning and Zhu (2012) 28 bps – Bonis, Ihrig, and Wei (2017)	22 bps	3.3 bps
QE3 ($790bn)	60 bps – Engen, Laubach, and Reifschneider (2015) 31 bps – Bonis, Ihrig, and Wei (2017)	45.5 bps	5.8 bps
Total		128.5 bps	6.35 bps

Return

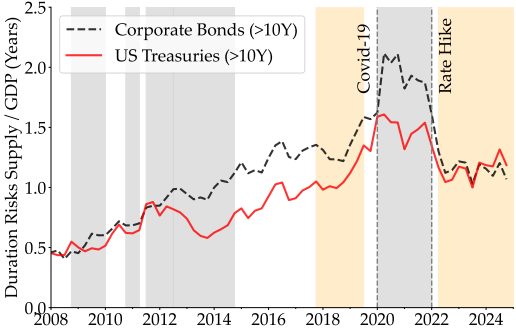
Treasury Market is less elastic when Allocators are more constrained

VARIABLES	Shock 1 (1) $\Delta y_{10\text{ min}}$	Shock 2 (2) $\Delta y_{10\text{ min}}$	Shock 3 (3) $\Delta y_{10\text{ min}}$
$Shock_t$	0.187**	7.811***	-1.369**
	(0.0734)	(2.890)	(0.531)
$Shock_t \times \Delta Treasury\ Share_t$	0.320**	14.66*	-0.414*
	(0.151)	(8.594)	(0.180)
$\Delta Treasury\ Share_t$	-0.214*	-0.00848	0.883
	(0.123)	(0.123)	(0.495)
Observations	433	1,013	1,233
R-squared	0.0887	0.057	0.095
Maturity FE	Y	Y	Y

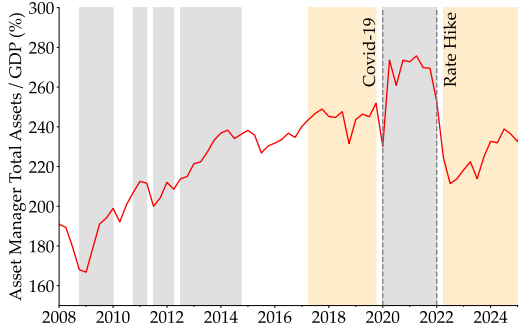
Return

Level of Duration Supply and Demand

(a) Duration supply



(b) Total asset of institutional bond investors



- Duration supply $= \frac{\text{Outstanding bond amount} \times \text{Weighted average duration}}{GDP}$.
- Exclude Treasuries held by the Federal Reserves.

Return

Robustness Check on the Shocks

1. Do markets know the shocks?
   Yes. Live news coverage of Treasury announcements from Bloomberg and Reuters.

2. Forecastable component in the shocks such that the shocks are overestimated?
   - Exclude quarters with forecastable fiscal expansion: 2008Q4, 2020Q2.
   - Exclude predictable components from Shock 1: changes in the previous-quarter cash balance.
   - Weak predictability of shocks using macro-financial variables: *GDP growth, unemployment rate, stock return, short-term rate, long-term rate, yield curve slope...*
   - Identification is valid if any bias in the shocks is not correlated with investors' constraint levels.

Result

Return to shocks

Predictability of Shock 1

VARIABLES	$\frac{\widetilde{\text{Borr. Est.}_t}}{D_t}$ (%)
$\Delta \log\ GDP_{t-1}$	0.0293
	(0.116)
Δ *Unemp. rate*$_{t-1}$	0.00355*
	(0.00197)
$\Delta \log\ StockIndex_{t-1}$	-0.0136**
	(0.00579)
Δ *FFER*$_{t-1}$	-0.000146
	(0.000302)
Δ 10*Y Treasury*$_{t-1}$	-6.10e-05
	(0.00160)
Δ *Slope*$_{t-1}$	-0.000463
	(0.00236)
Constant	0.000140
	(0.00154)
Observations	126
R-squared	0.141
Corr(*Shock*, *Residual*)	0.927

Return

Composition of US Long-term Debt Market

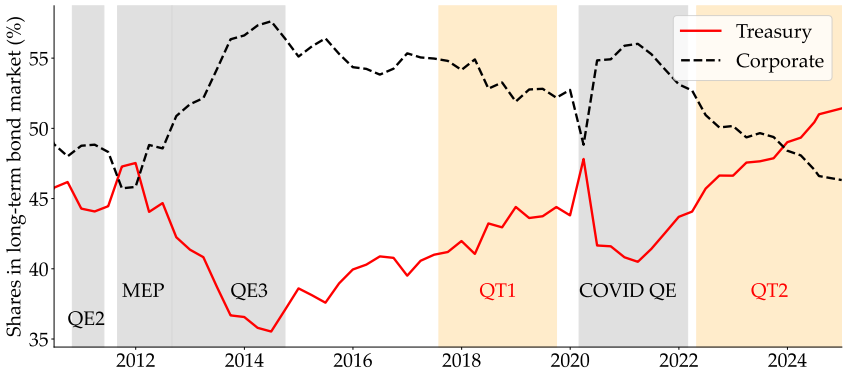
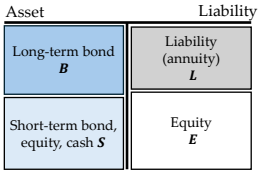


Figure: The Treasury share and corporate bonds share in the long-term bond market

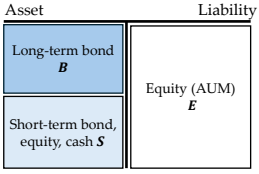
Return

Balance Sheet of Liability Driven Investors

(a) Liability driven investor



(b) Bond mutual funds



- LDI investors: immunize equity value $E = B + S - L$ from interest rate changes.

$$w_B \equiv \frac{B}{L} = \frac{D_L}{D_B}$$

- Bond mutual funds: match duration of benchmark index D_I.

$$w_B \equiv \frac{D_I}{D_B}$$

Return

Treasury Market Elasticity in the Simplified Model

Proposition

Consider the economy with perfectly elastic short-term corporate bond market. Comparing with the slack region, a marginal Treasury supply shock has

i) *a stronger price impact on Treasury yield under the binding region:*

$$\frac{\partial y_0^T}{\partial S_T} = \underbrace{\frac{1}{2}\frac{1}{\varepsilon_T + \varepsilon_L}}_{\text{slack region}} < \underbrace{\frac{1}{2}\frac{1}{\varepsilon_T + \varepsilon_L \frac{\varepsilon_A}{\varepsilon_A + \varepsilon_L}}}_{\text{binding region}}, \quad \text{with } \varepsilon_A \equiv \frac{1}{\gamma \sigma_z^2}$$

ii) *a weaker price impact on long-term corporate bond yield under the binding region:*

$$\frac{\partial y_0^L}{\partial S_T} = \underbrace{\frac{1}{2}\frac{1}{\varepsilon_T + \varepsilon_L}}_{\text{slack region}} > \underbrace{\frac{1}{2}\frac{1}{\varepsilon_T \frac{\varepsilon_A + \varepsilon_L}{\varepsilon_A} + \varepsilon_L}}_{\text{binding region}}, \quad \text{with } \varepsilon_A \equiv \frac{1}{\gamma \sigma_z^2}$$

Return

Treasury Market Elasticity in the Full Model

The economy features two equilibria region, with

i) *less* elastic Treasury market under the *binding region*:

$$\varepsilon_{TT}: \quad \underbrace{2\left(\varepsilon_T + \varepsilon_L \frac{\varepsilon_A + \varepsilon_S}{\varepsilon_A + \varepsilon_L + \varepsilon_S}\right)}_{\text{slack region}} > \underbrace{2\left(\varepsilon_T + \varepsilon_L \frac{\varepsilon_A}{\varepsilon_A + \varepsilon_L}\right)}_{\text{binding region}} \tag{1}$$

ii) *weaker* price impact on long-term corporate bond yields under the *binding region*:

$$\frac{\partial y_0^L}{\partial S_T} = \underbrace{\frac{1}{2}\frac{1}{\varepsilon_T \frac{\varepsilon_A+\varepsilon_L+\varepsilon_S}{\varepsilon_A+\varepsilon_S} + \varepsilon_L}}_{\text{slack region}} > \underbrace{\frac{1}{2}\frac{1}{\varepsilon_T \frac{\varepsilon_A+\varepsilon_L}{\varepsilon_A} + \varepsilon_L}}_{\text{binding region}} \tag{2}$$

Specifically, when the market is fully segmented, $\varepsilon_{TT} = \varepsilon_T$. When the market is non-segmented, $\frac{\partial y_0^L}{\partial S_T} = 0$.

Return

Treasury Market Elasticity with Basis Trade

Treasury market elasticity:

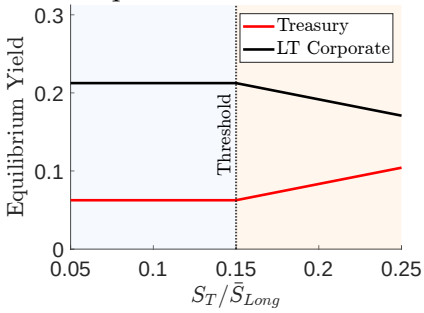
$$\varepsilon_{TT} : \underbrace{2\left(\varepsilon_T + \varepsilon_L \frac{\varepsilon_A + \varepsilon_S}{\varepsilon_A + \varepsilon_L + \varepsilon_S}\right)}_{\text{slack region}, \equiv \varepsilon_{TT}^{\text{Slack}}} > \underbrace{2\left(\varepsilon_T + \varepsilon_L \frac{\varepsilon_A\left(1 + \frac{\varepsilon_S}{\varepsilon_D}\right) + \varepsilon_S}{(\varepsilon_A + \varepsilon_L)\left(1 + \frac{\varepsilon_S}{\varepsilon_D}\right) + \varepsilon_S}\right)}_{\text{binding region + basis trade}, \equiv \varepsilon_{TT}^{\text{BasisTrade}}} > \underbrace{2\left(\varepsilon_T + \varepsilon_L \frac{\varepsilon_A}{\varepsilon_A + \varepsilon_L}\right)}_{\text{binding region}, \equiv \varepsilon_{TT}^{\text{Binding}}}$$

Return

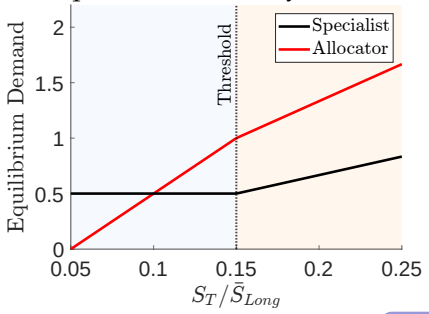
Comparative Analysis

Holding total duration risk $S_L + S_T$ and credit risk $S_L + S_S$ fixed

(a) Equilibrium Bond Yields



(b) Equilibrium Treasury Demands



Return

Threshold Condition in the Full Model

The threshold condition for the *slack region* and *binding region* follows

$$g(S_T, S_L, S_S, \bar{D}) \equiv \frac{\varepsilon_T \varepsilon_S}{\Delta} S_L - \frac{\varepsilon_L \varepsilon_S}{\Delta} S_T - \frac{\varepsilon_A(\varepsilon_L + \varepsilon_T) + \varepsilon_T \varepsilon_L}{\Delta} S_S + 1 - \frac{\varepsilon_A(\varepsilon_L + \varepsilon_T) + \varepsilon_T(\varepsilon_L + \varepsilon_S)}{\Delta} \bar{D}$$

$$\begin{cases} \geq 0 & \text{(slack region)}, \\ < 0 & \text{(binding region)}, \end{cases} \qquad \text{where} \quad \Delta \equiv (\varepsilon_S + \varepsilon_A)(\varepsilon_L + \varepsilon_T) + \varepsilon_T \varepsilon_L$$

Comparative analysis:

1. Holding total duration risk $S_L + S_T$ and total credit risk $S_L + S_S$ fixed. A lower S_L increases the likelihood of entering the binding region.
2. Allocators with a higher duration mandate $\bar{D}$ are more likely to bind.

Return

Gap Between Long-term and Short-term Credit Spread

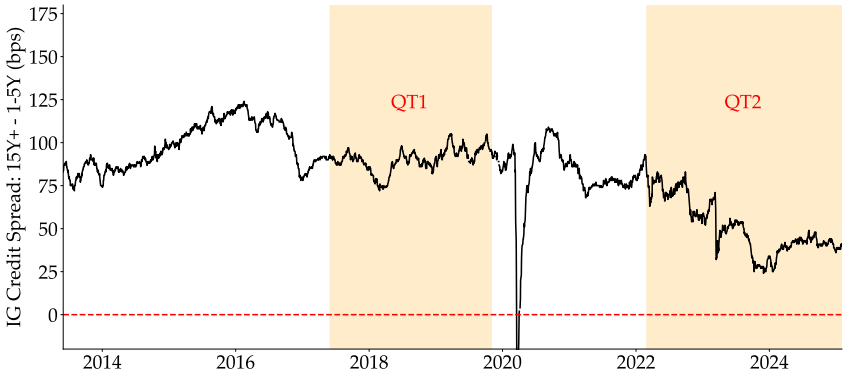


Figure: US Corporate Bond Credit Spread: 15+ Year - 1-5 Year.

Return to model

Limits of Treasury Market Elasticity

- Elasticity restored "*partially*" due to derivative basis charged by hedge funds.
- if hedge funds have perfect elastic supply = leverage freely ⇒ slack region

$$\lim_{\varepsilon_D \to \infty} \varepsilon_{TT}^{\text{BasisTrade}} = \varepsilon_{TT}^{\text{Slack}}$$

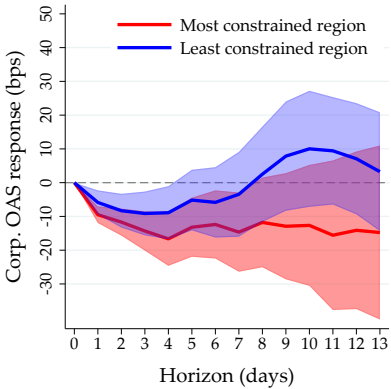
- if hedge funds have inelastic supply = no-leverage ⇒ binding region

$$\lim_{\varepsilon_D \to 0} \varepsilon_{TT}^{\text{BasisTrade}} = \varepsilon_{TT}^{\text{Binding}}$$

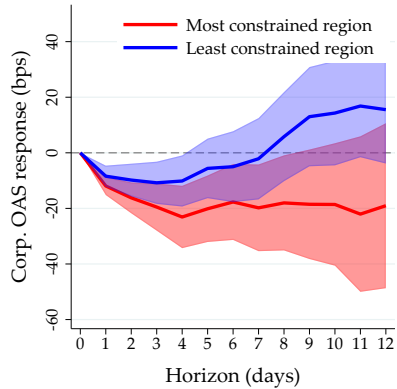
Return

Pass-through to Lower Rating Bonds on Auction Days

(a) All Investment-grade bond



(a) BBB corporate bond



Return

Pass-through on Long-term Treasury Auction Days

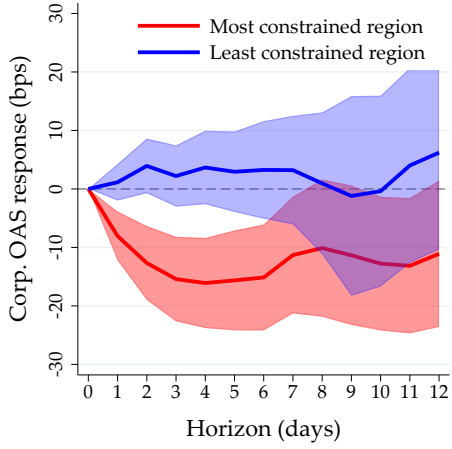


Figure: Long-term Treasury auctions only

Swap Spread and CDS Basis

	Treasuries	Corporate Bond
Cash Bond	Treasury yield	Credit spread
Derivative	Swap rate	Credit default swap
Difference	Swap Spread	CDS Basis

- Typically:
  - CDS Basis > 0: funding cost and liquidity cost associated with holding cash bonds.
  - Swap spread $= 0$: equivalence of cash Treasuries and Treasury derivatives
- Now: CDS Basis $\rightarrow 0$; Swap Spread $\rightarrow$ -20 bps.
- Funding cost of holding cash Treasury $\uparrow$; holding cash corporate bond $\downarrow$.

Return

Pass-through to Corporate Bond Market: High Frequency Shocks

	(1)	(2)	(3)
VARIABLES	$h=1$	$h=2$	$h=3$
$\Delta y_t^{10Y,10min}$	-6.018	-33.58*	-34.74*
	(12.05)	(18.63)	(20.48)
$\Delta y_t^{10Y,10min} \times Q_{Swap\ Spread_t^{10Y}} = 2$	8.929	52.18**	54.39*
	(14.60)	(24.06)	(28.86)
$\Delta y_t^{10Y,10min} \times Q_{Swap\ Spread_t^{10Y}} = 3$	6.566	14.22	7.498
	(13.79)	(21.73)	(25.86)
$\Delta y_t^{10Y,10min} \times Q_{Swap\ Spread_t^{10Y}} = 4$	1.087	42.31**	45.82*
	(13.56)	(21.44)	(26.07)
$Q_{Swap\ Spread_t^{10Y}} = 2$	0.161	0.437	0.470
	(0.114)	(0.287)	(0.345)
$Q_{Swap\ Spread_t^{10Y}} = 3$	0.265**	0.560*	0.708*
	(0.113)	(0.318)	(0.399)
$Q_{Swap\ Spread_t^{10Y}} = 4$	0.244**	0.512*	0.459
	(0.105)	(0.280)	(0.349)
Observations	1,035	1,035	1,035
R-squared	0.056	0.141	0.134
Controls	Y	Y	Y

$$\Delta OAS_{t-1\to t+h}^{AAA} = \alpha_h + \beta_h \Delta y_t^{10Y,10\text{min}} + \sum_{j=1}^{3} \Delta OAS_{t-j-1\to t-j} + \text{Controls}_t + \varepsilon_{t,h}$$

Return

Validating the Derivative Spread Measure

1. **Supply side reason: time-varying per-unit derivative intermediation costs**
   - Limit sample to post-2009 period when comparing Treasury supply effects
   - VAR decomposition confirms swap spreads are demand-side driven

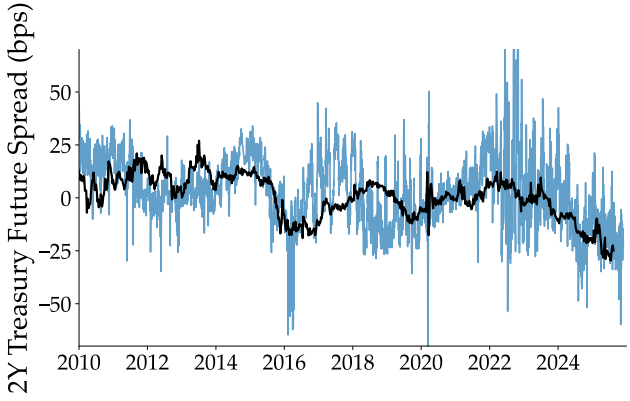
◂ VAR Decomposition

2. **Other market participants influence swap spreads**
   - Futures swap spreads comove strongly with spot swap spreads

◂ Future Spread

◂ Return to Model Summary

Treasury Future Spread Correlated with Swap Spread

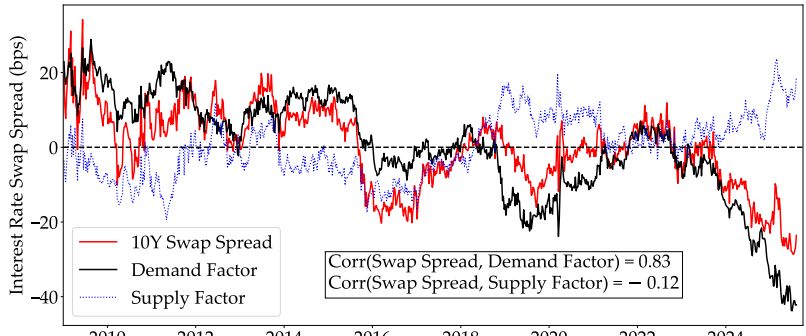


$$\text{Future spread} = \underbrace{\text{term repo rate}}_{\text{funding cost of holding cash Treasury}} - \underbrace{\text{implied repo rate}}_{\text{funding cost of holding Treasury future}}$$

Future spread < 0: Treasury future is overpriced. Long cash bond and short future.

Return

VAR Decomposition of Swap Spread Hanson et al. (2024)

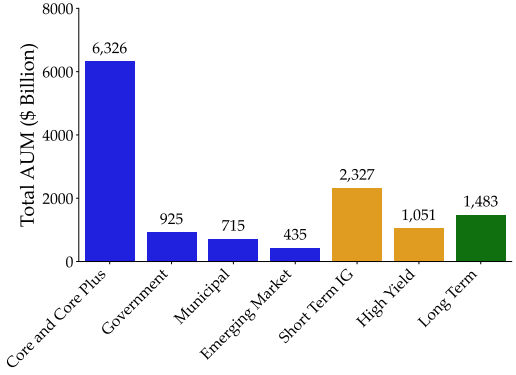


- Demand side: low swap spread due to high end-user demand
  - → high intermediation quantity.
- Supply side: low swap spread due to high HF/dealer per unit intermediation cost
  - → low intermediation quantity.

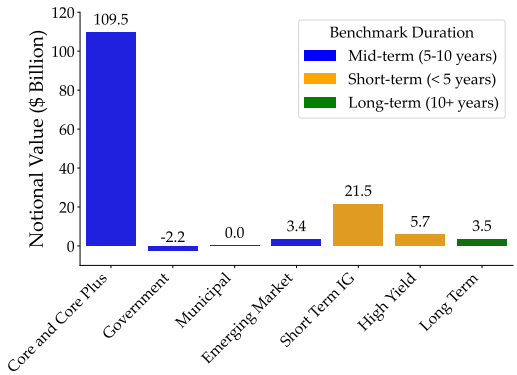
Return

Bond Mutual Funds: Who Uses Derivatives?

(a) Moringstar Global Bond Fund AUM (2024Q1)



(b) Moringstar Bond Fund Treasury Future Positions (2024Q1)



- Core and Core Plus funds are primarily benchmarked to the Bloomberg US Aggregate Bond Index (or Bloomberg US Universal Bond Index).

Return

Bond Mutual Funds: Index Benchmarking

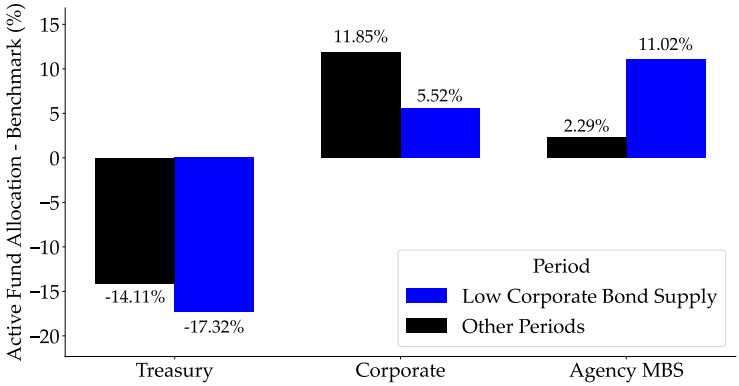
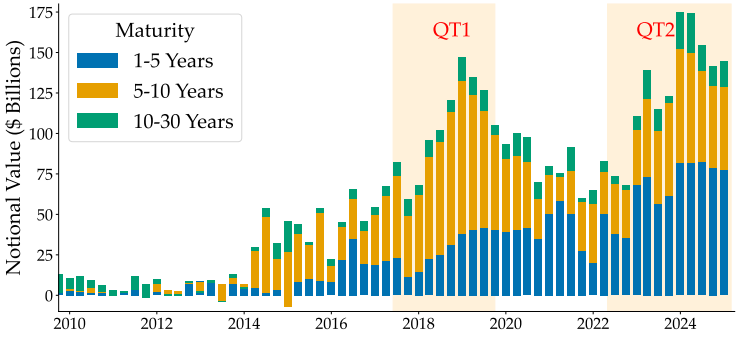


Figure: Active-managed fund asset allocation relative to Bloomberg benchmark index

Return to Anecdotal Evidence Return to Facts

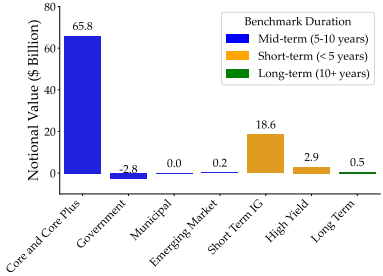
Bond Mutual Funds Treasury Future Usage



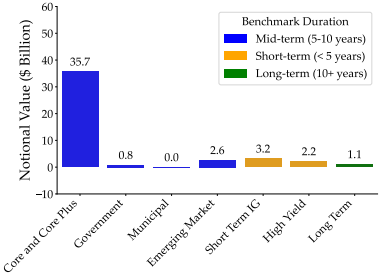
Return

Bond Mutual Funds: Future Positions by Tenor

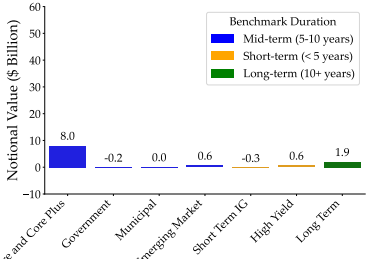
(a) 1-5 Y



(b) 5-10Y



(c) 10Y +



Return

Life Insurance Company Duration

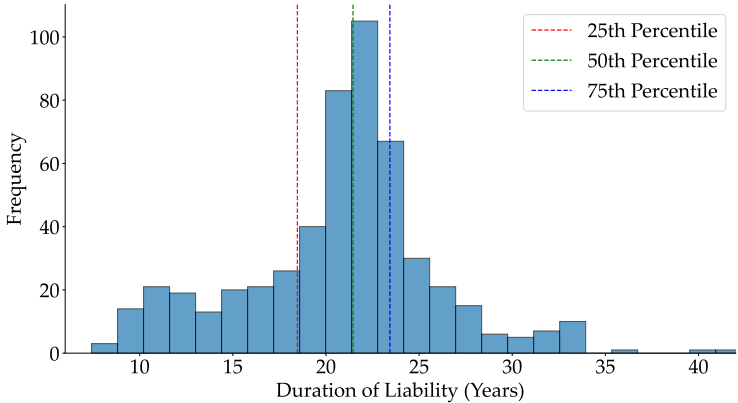
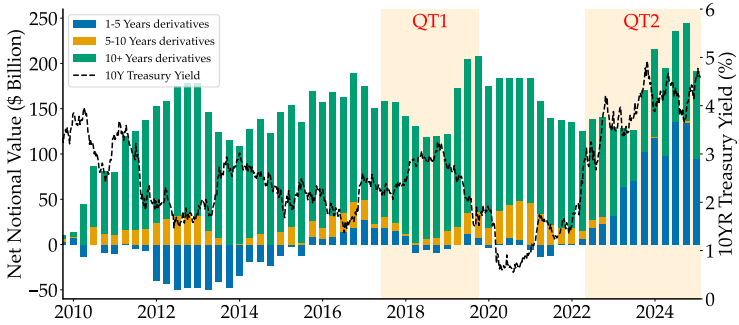


Figure: Distribution of US Life Insurance Company Liability Duration (2020)

Return

Life Insurance Treasury Derivative Usage



Return